Plan — R11.3: Coin/Target Demonstration-Bank Pilot

Certificate-gated demonstration *generation* (no BC/RL/refinement): one teacher-solved rollout per coin/target scenario, from the exact zero home through strict K6 or a *classified* failure, with full provenance + measured energy ledger. Pilot first (∼12 admissible scenarios + an invalid panel), then **HALT** for review.

HyMeKo coin-delivery campaign (`feature/r11-3-coin-target-demo-bank`, from R11.2 `fb5b43a2`)

2026-07-29

Abstract

R11.2 (committed) built the HyMeKo IR + initial-condition / energy / provenance certificates. R11.3 uses them to *generate demonstrations*: for each coin/target scenario, run the certified pipeline EXACT_ZERO_HOME → IC certificate → admissibility → **deployed** RRT reach → precontact handoff → **per-instance CEM capture/delivery teacher** → strict K6 or a classified failure, and record everything (IC cert, sample ID, admissibility, RRT path + reach cert, coin/target poses, relative goal vector, handoff descriptor, mode trace M0–M7, teacher params, actions/observations, contact sequence, K6 result, failure class, measured energy ledger, energy-transition certificates, provenance + content hash). **This boundary is generation only** — no BC, no RL, no policy assimilation, no geometry refinement, no transition optimization, no Hamiltonian reward shaping. A *pilot* (∼12 admissible scenarios + a separate invalid-initial-condition panel) proves the pipeline before the full bank is generated. Verdict target: `R11_3_COIN_TARGET_DEMONSTRATION_PIPELINE_PASS`.

1 Scope, non-goals, boundary

Overnight extension (2026-07-30). This plan's initial failure taxonomy was extended to **14 classes** (adding `GOAL_SET_EMPTY`, `PREMATURE_CONTACT`, `PRECONTACT_HANDOFF_INVALID` vs. `CAPTURE_/DELIVERY_TEACHER_FAILURE`, `PROVENANCE_FAILURE`, `ENERGY_LEDGER_INCOMPLETE`) to separate the R11.3 teacher-generation-contract gap (reach lands outside the teacher's admissible handoff manifold) from downstream capture/delivery failures; the pilot was gated (14 checks) and, on PASS, a bounded bank of 64 scenarios × ≤ 3 teacher seeds (≤ 192 attempts) was generated with a coverage analysis. See the report for the final state.

Goal. A reusable, certificate-gated demonstration generator + on-disk bank, validated by a small pilot. **Roles (unchanged from R11.2).** RRT-Connect is the *deployed* geometric planner (reach). The **CEM capture/delivery is a training teacher only**; every teacher-produced trajectory is labelled as such in provenance and is *never* presented as teacher-free deployment. **Hard non-goals for R11.3:** no BC, no RL, no policy assimilation, no geometry/timing refinement, no transition optimization, no Hamiltonian reward shaping, no energy *optimization* (the ledger is measured, per the R11.2 two-level contract). **Stop after the pilot report** for review; the full bank is generated only after the pilot verdict.

2 Per-scenario pipeline

For each scenario (x_c, x_g):

1. Build EXACT_ZERO_HOME; `certify_zero_home` ⇒ IC certificate (must be valid, else the rollout is void, not a failure sample).

2. `coin_admissibility`(x_c) + target-in-bounds. **Inadmissible ⇒ `INVALID_INITIAL_CONDITION`: rejected *before* planning, ledgered in the rejection panel, and *excluded from the success denominator*.**
3. Deployed RRT reach (from R11.1) ⇒ reach certificate (path + clearance).
4. Precontact handoff ⇒ `HandoffDescriptor` (complete).
5. Per-instance CEM capture/delivery teacher (re-solved for this scenario) toward the target zone.
6. Strict-K6 evaluation ⇒ success *or* a specific `FailureClass`.
7. Emit provenance (with content hash) + measured energy ledger + energy-transition certificates.

See `plan.tikz` for the dataflow and `plan.mmd` for the decision/failure branches.

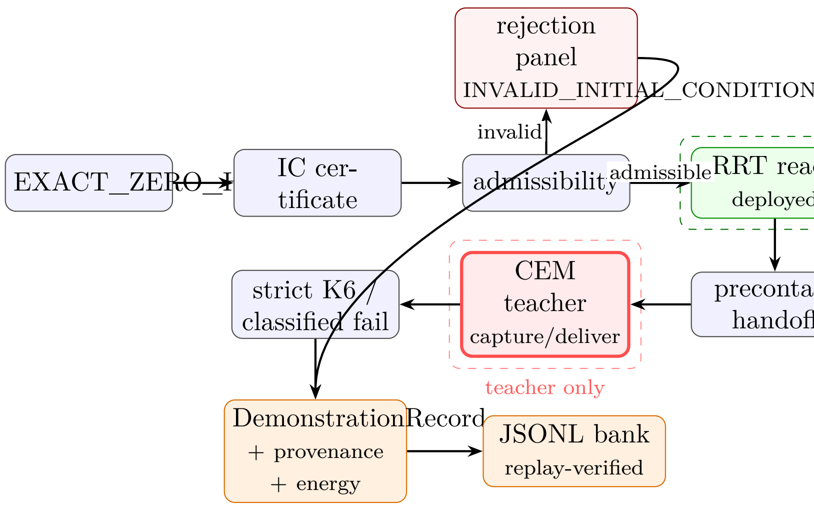

Figure 1: The R11.3 per-scenario demonstration pipeline. Green = the deployed geometric planner (RRT reach); red = the CEM *teacher* (training only, labelled in provenance); every stage emits into the record, which is serialized to the JSONL bank. An inadmissible start is rejected *before* planning into the separate rejection panel (never in the success denominator); any downstream stage can terminate in a specific `FailureClass`.

3 Scenario design (disjoint IDs; no interpolation leakage)

The **target** is the delivery zone (`_zone_x`, `_zone_y`); the **relative goal vector** is $\Delta x_g = x_g - x_c$. Pilot set (∼12 admissible + an invalid panel), each a stable scenario ID assigned to a `train/dev/test` split (disjoint):

- canonical coin + canonical target;
- coin + target translated together (relative vector fixed — G0);
- shifted coin, canonical relative target vector (G0/G2);
- fixed coin, changed target direction (G1) — realized by relocating the zone where the sim honors it, else a classified delivery/target-entry failure (an honest teacher-capability datum, not a hidden gap);
- ≥ 1 diagonal coin/target case;
- ≥ 1 invalid-initial-condition case (e.g. `right +5cm`) in the separate rejection panel.

4 Storage schema (one record per rollout)

`DemonstrationRecord`: IC certificate; distribution sample ID; admissibility result + rejection reason; RRT path + reach certificate; coin pose; target pose; relative goal vector; precontact handoff descriptor; mode trace M0–M7; teacher capture + delivery parameters; actions; observations; contact sequence;

strict-K6 result; failure classification; measured energy ledger; energy-transition certificates; rollout provenance + content hash. Serialized to JSONL; a `replay` reconstructs the scenario + result and must reproduce the recorded content hash.

5 Failure taxonomy (enum — never collapsed to one label)

`FailureClass`: `INVALID_INITIAL_CONDITION`, `REACH_PLANNER_FAILURE`, `REACH_EXECUTION_FAILURE`, `PREMATURE_COIN_MOTIO` `CAPTURE_FAILURE`, `DELIVERY_FAILURE`, `TARGET_ENTRY_OVERSHOOT`, `SETTLE_K6_FAILURE`, `SAFETY_FAILURE`. Each rollout ends in exactly one of {success, one of these}.

6 Affected files (new; branch from the R11.2 commit; no CORE.YAML items)

`hymeko_rl/coin_delivery/demo_bank/` (`scenario.py` coin/target scenario + disjoint grid; `failure_class.py` the enum + `classify`; `record.py` `DemonstrationRecord` + serialize/hash/replay; `pipeline.py` `run_scenario` (the certified pipeline, reusing `ir_adapter` + RRT + CEM teacher); `store.py` `DemonstrationBank` JSONL read/write). `hymeko_rl/experiments/r11_3_demo_pilot.py` (pilot runner). `hymeko_rl/tests/test_r11_3_demo_bank.p` (pilot gates). Reuse `hymeko_rl.ir`, `coin_delivery.ir_adapter`, `coin_zero_home_rrt`, the frozen CEM teacher + downstream.

7 Pilot gates

`EVERY_ROLLOUT_FROM_VALID_EXACT_ZERO_CERTIFICATE`, `DISJOINT_TRAIN_DEV_TEST_SCENARIO_IDS`, `INVALID_STARTS_REJEC` `NO_CEM_USE_HIDDEN_FROM_PROVENANCE`, `EACH_SUCCESS_LINKS_K6_TO_PROVENANCE_HASH`, `EACH_ROLLOUT_HAS_COMPLETE_MODE` `EACH_ROLLOUT_HAS_COMPLETE_MEASURED_ENERGY_LEDGER`, `SERIALIZATION_REPLAY_REPRODUCES_SCENARIO_AND_RESULT`, `NO_TEACHER_TRAJECTORY_LABELLED_TEACHER_FREE`.

8 Test / perf / risk / rollback

Tests: unit (scenario grid disjointness, failure `classify`, record serialize/hash/replay round-trip, rejection accounting) fast; integration (one full pilot rollout: canonical success + one classified failure + one rejected invalid) slow. **Determinism**: fixed seeds; RRT seed pinned; content hash float-rounded to 1e-9. **Perf**: pilot ~12 scenarios × (reach + CEM capture ≈ the R11.2 slow-test cost per instance); reconcile the wall estimate (> 2× rule) before the run; well under the 16 GB RSS cap (CPU, no training). **Risk**: (i) shifted-coin / changed-target scenarios may not deliver K6 with the canonical teacher — that is a *classified failure*, recorded honestly, never an impossibility claim; (ii) zone relocation may not be honored by the downstream — if so, target-variation scenarios classify as delivery/target-entry failures and this is stated in the pilot report; (iii) leakage — train/dev/test IDs disjoint by construction ([[held-out panel is single-use]]). **Rollback**: new package + experiment + test; rollback = delete. No existing file modified.

9 Stop logic

Run the pilot, verify the nine gates, write the pilot report, and **HALT**. **No BC/RL/refinement, and no full-bank generation, until the pilot verdict `R11_3_COIN_TARGET_DEMONSTRATION_PIPELINE_PASS` is reviewed.**